

Where revenue hides losses

A cleaned order export, a profitability review and a prioritised set of business questions.

\$2.30m

historical revenue

\$286.4k

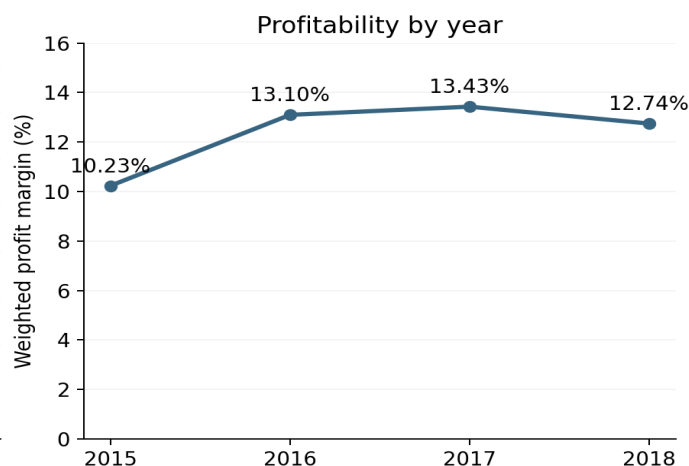
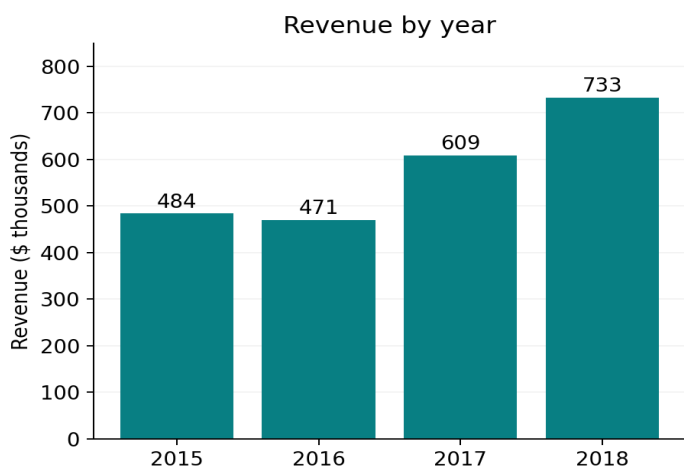
historical profit

12.47%

weighted profit margin

Business question

Which parts of the business deserve closer review before increasing discounts or targeting customers?
This analysis turns a historical Superstore-style sample into a reproducible decision brief.



Data quality, with an audit trail

Cleaning stage	Order lines
Raw input	10,800
Exact duplicates removed	504
Invalid rows removed after deduplication	302
Valid lines retained	9,994

5,009 distinct orders | 793 customers | 3 Jan 2015 - 30 Dec 2018

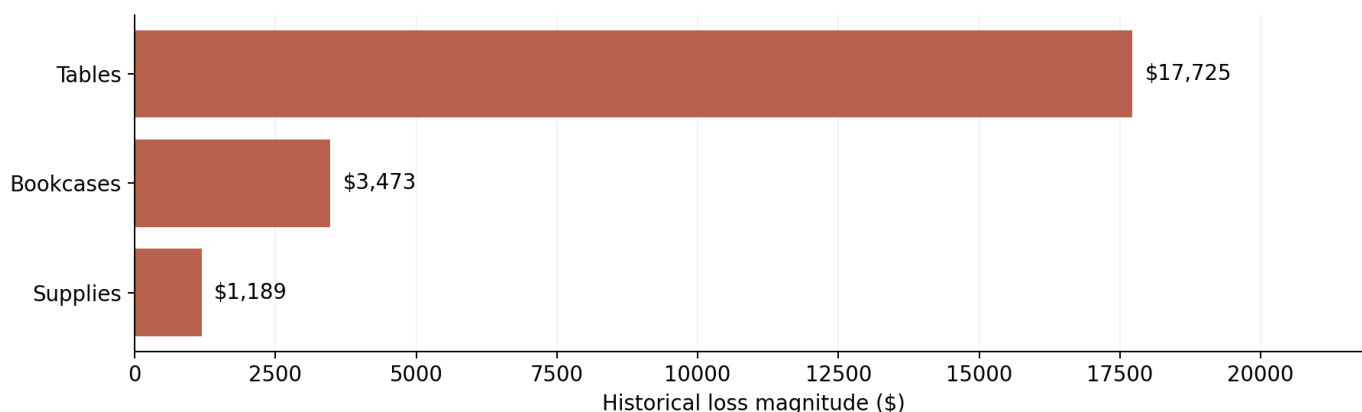
Revenue and profit use the values in the sample; dollar labels assume USD. These are historical sample totals, not revenue earned or improvements delivered to a client.

Three decisions to investigate

Recommendations follow the observed data; they do not claim a causal impact or guaranteed return.

1. Review loss-making product groups

Tables, Bookcases and Supplies together show a \$22,387 historical loss. Tables account for \$17,725 of that total. Check product cost, shipping allocation and discount rules before proposing changes.



2. Test discount rules rather than assume causality

Every observed discount band above 20% has negative average line profit. The sample also differs by product and order mix. Review comparable products and run a controlled pricing test before asserting that a discount cap will improve profit.

3. Plan a measurable retention pilot

The RFM snapshot identifies 146 customers in the heuristic At Risk segment and 167 Champions. These labels combine recency and frequency; monetary value is reported separately. Validate segment definitions and campaign eligibility, then use a holdout group to measure incremental impact.

Client-ready delivery

Deliverable	Purpose
Cleaning audit + source hash	Trace every exclusion and reproduce totals
Trend, category, discount and RFM tables	Review the underlying evidence
Report + figures + Python script	Repeat the work on a future export

Reproduce: `python sales_analysis.py`; `python -m pytest -q`; `python portfolio_figures.py`. The three sales tests passed. Original exploratory charts remain separate from these current figures.

Source: `data/superstore.csv`, historical Superstore-style sample bundled in the repository. The original external source and licence are not documented in the existing project; no real-client provenance is claimed. Input SHA256 begins `75264adc0cf180c4`; the full value and cleaning rules are in `reports/client/`.